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(54) **COMPUTER CASE WITH GRAPHICS CARD HOLDER**

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G06F 1/18 (2006.01)

(52) **U.S. Cl.**
CPC ***H05K 7/1461*** (2013.01); ***G06F 1/183*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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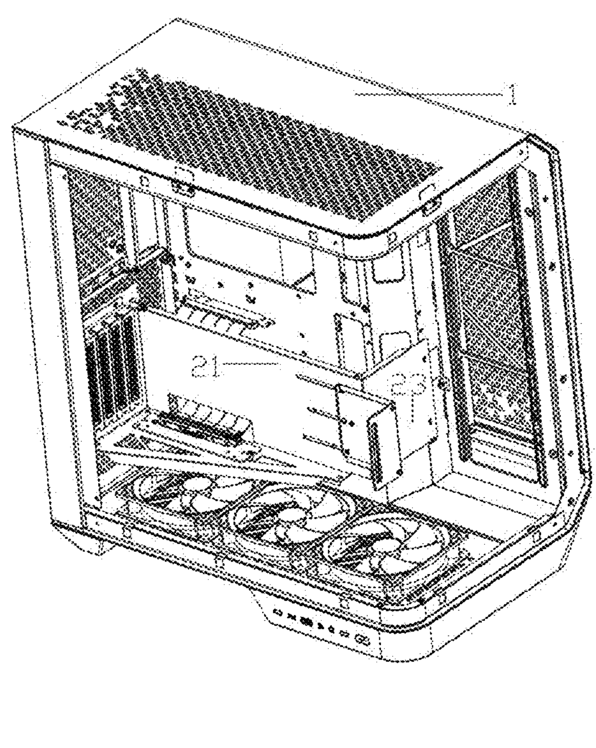
(57) **ABSTRACT**

A computer case with a graphics card holder is provided, which includes a computer case body and a graphics card holder. The graphics card holder includes a first support member, a second support member, and a third support member. The first support member is slidably connected to a fixing member, the second support member is vertically provided at one end of the first support member, the second support member is provided with a graphics card adapter cable. One side of the third support member away from the first support member is fixedly connected to the computer case body. By setting the graphics card holder and fixing component, the computer case can stably fix the graphics card. The setting of the graphics card adapter cable allows the graphics card to be placed vertically, thereby enhancing the heat dissipation performance of the graphics card.

8 Claims, 6 Drawing Sheets



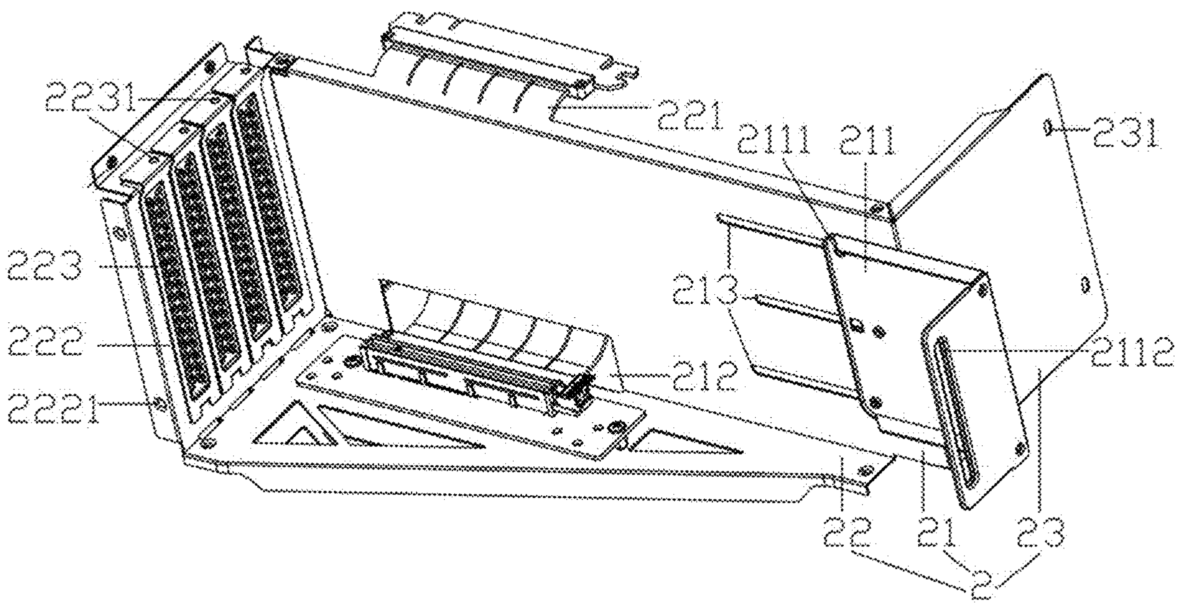


FIG. 1

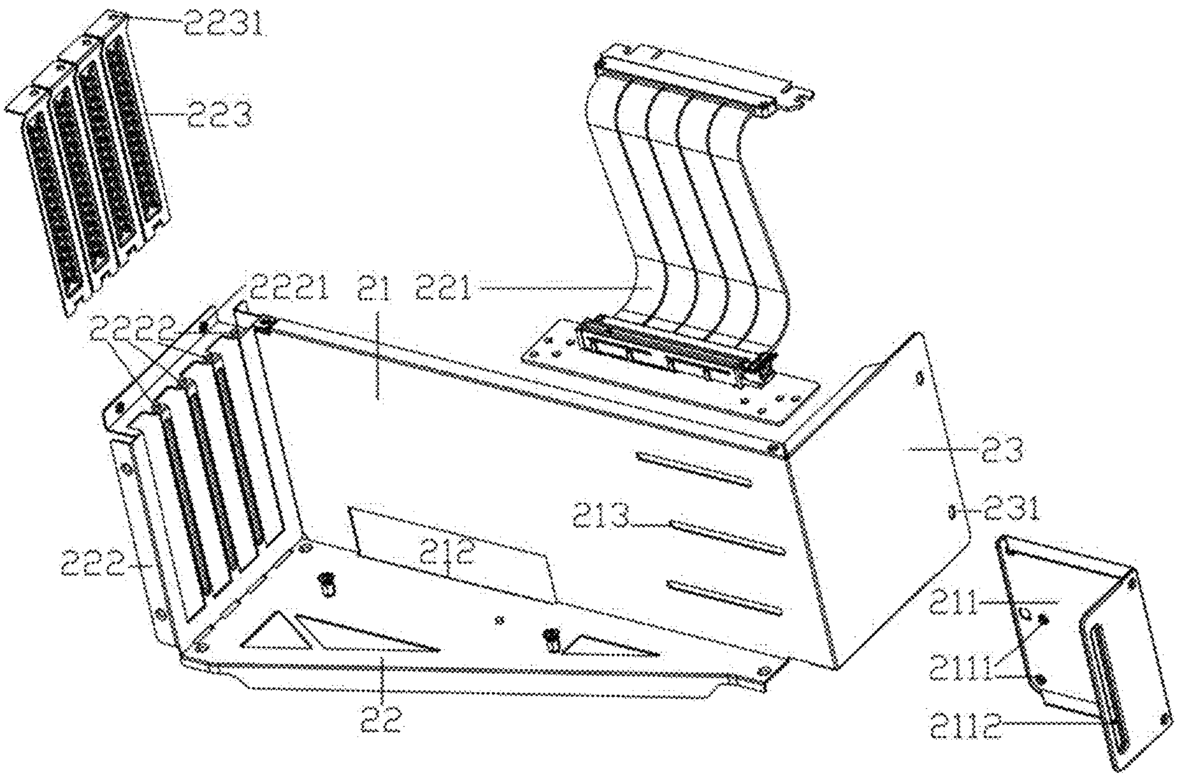


FIG. 2

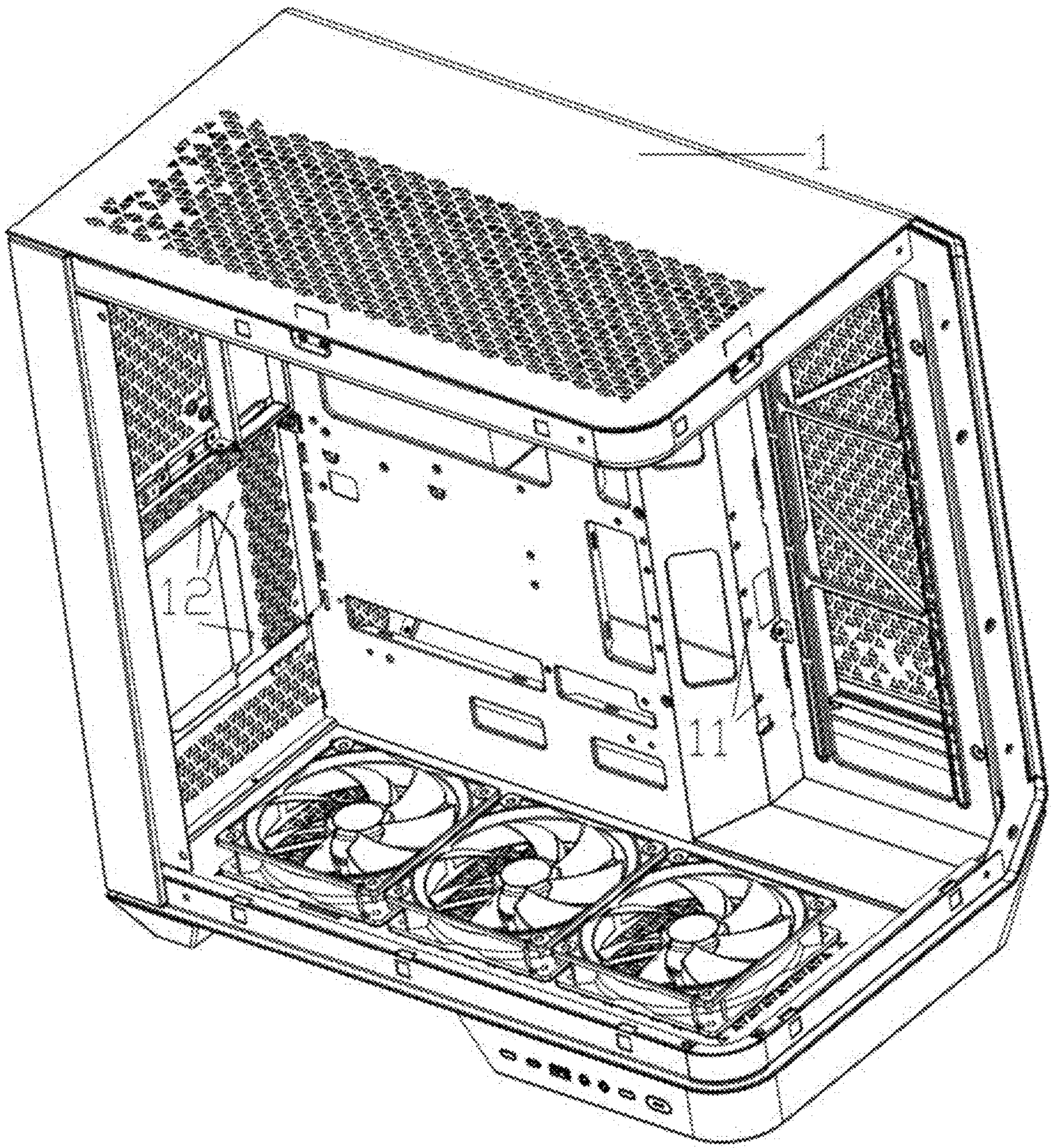


FIG. 3

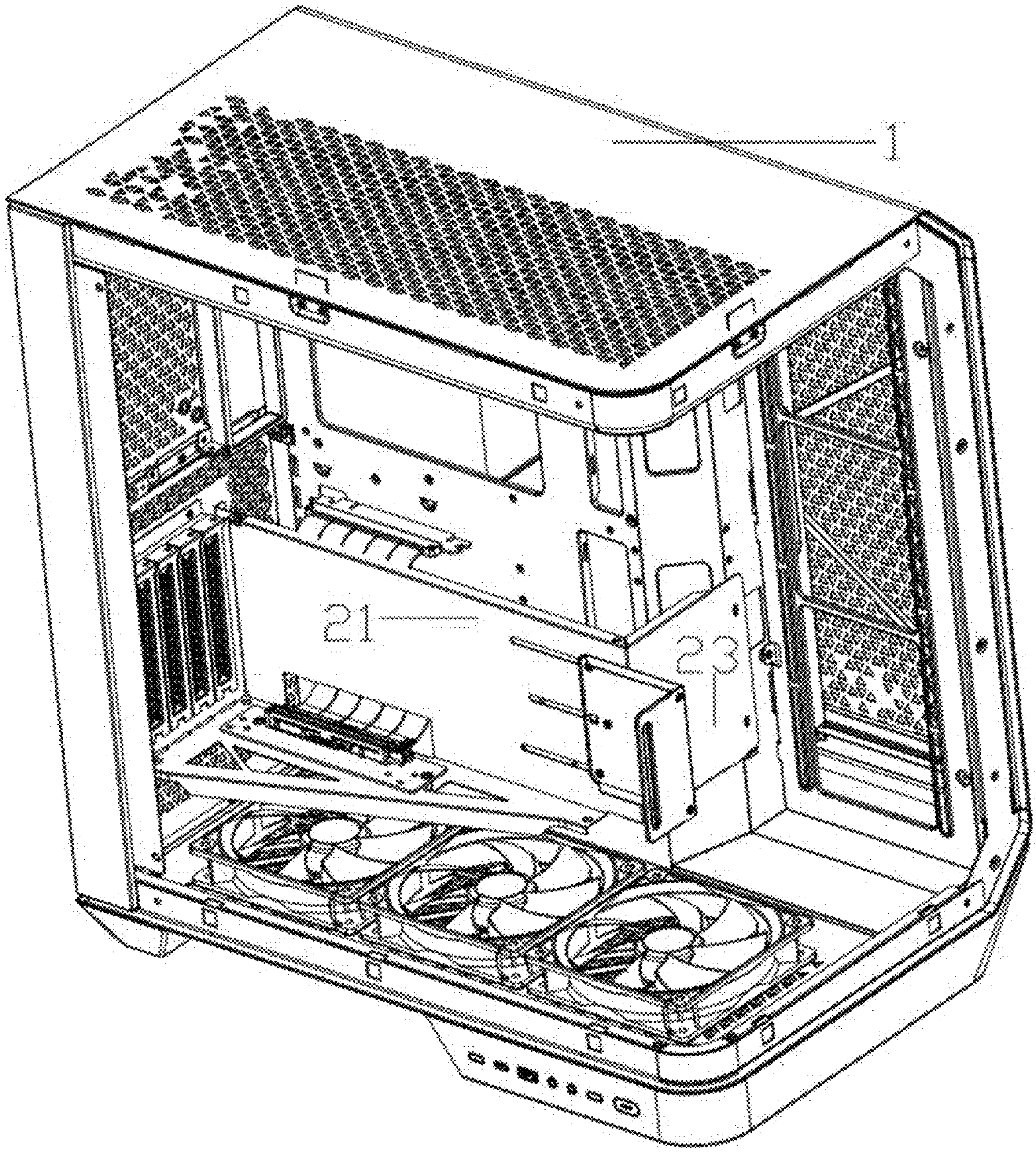


FIG. 4

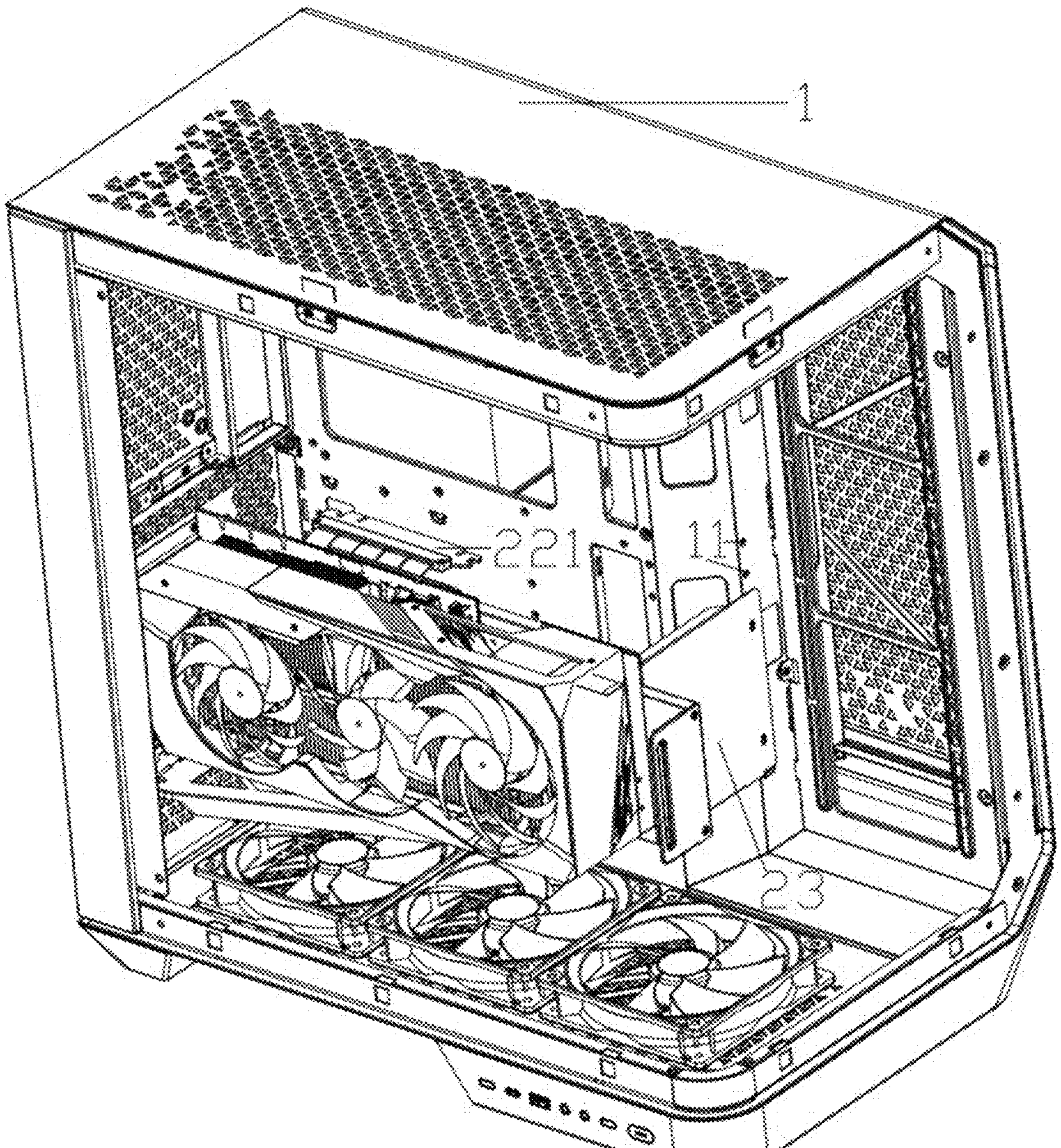


FIG. 5

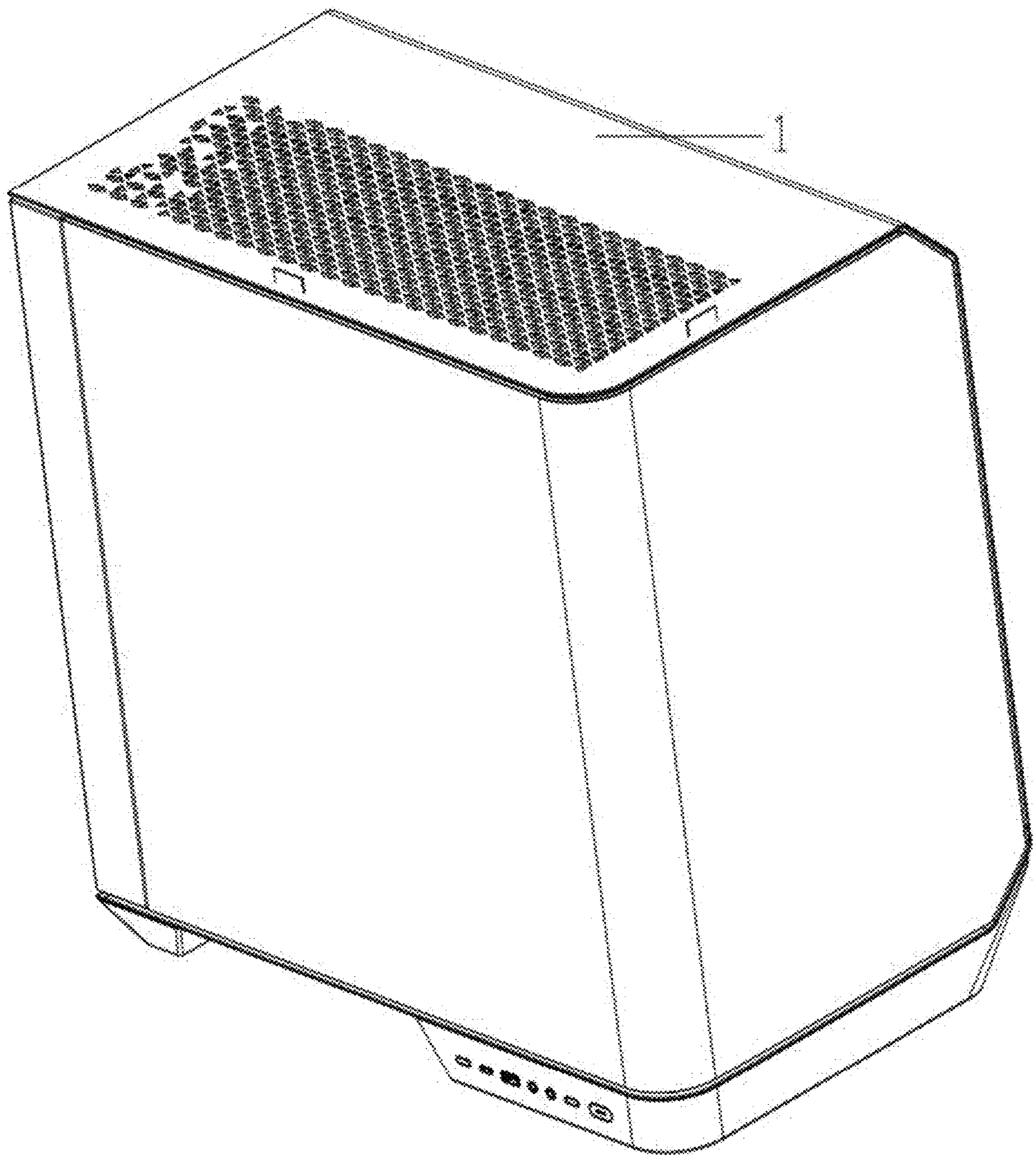


FIG. 6

COMPUTER CASE WITH GRAPHICS CARD HOLDER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to Chinese Patent Application No. 202420366050.0, filed on Feb. 28, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to the field of computer case products technologies, and in particular, to a computer case with a graphics card holder.

BACKGROUND

At present, conventional graphics cards are connected to the computer case motherboard in a horizontal manner, so they are basically placed horizontally, that is, the graphics card is vertically arranged with the computer case motherboard. The main disadvantage of horizontal placement is that the fan is placed at a bottom of the graphics card, and the hot air discharged by the fan floats upward while the fan blows downward, which causes the airflow to be obstructed and results in poor heat dissipation performance of the graphics card. In addition, the horizontal placement of the graphics card is unstable, and it is easy to bend and deform at a connection between the graphics card and the motherboard of the computer case, which affects the performance of the computer. Furthermore, because the fan is at the bottom, the colorful and cool fan cannot be visually seen, which reduces the visual beauty of the computer case.

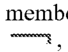
SUMMARY

To solve problems of poor heat dissipation performance, unstable graphics card fixation, and low visual aesthetics in most computer cases.

The present disclosure provides a computer case with a graphics card holder, including a computer case body and a graphics card holder, where the graphics card holder includes a first support member, a second support member, and a third support member; the first support member is slidably connected to a fixing member, the second support member is vertically provided at one end of the first support member, the second support member is provided with a graphics card adapter cable, the third support member is vertically provided on one side of the first support member, and one side of the third support member away from the first support member is fixedly connected to the computer case body.

In some embodiments of the present disclosure, the second support member is a right-angled trapezoidal structure, and one right-angled side of the second support member is connected to the first support member.

In some embodiments of the present disclosure, a threading port with a size larger than the graphics card adapter cable is provided on one side of the first support member near the second support member.

In some embodiments of the present disclosure, the fixing member is integrally formed and bent to form a shape of , at least two sliding holes are provided on one side of the fixing member, and a graphics card fixing hole is provided on the other side.

In some embodiments of the present disclosure, at least two parallel sliding grooves are provided on one side of the first support member near the third support member, and number and positions of the sliding grooves are corresponding to those of the sliding holes.

In some embodiments of the present disclosure, a distance between the first support member and a computer case motherboard is 4 cm to 8 cm.

In some embodiments of the present disclosure, one side of the third support member away from the first support member is provided with a first fixing hole, and the computer case body is provided with a second fixing hole corresponding to the first fixing hole.

In some embodiments of the present disclosure, one side of the second support member is bent to form a PCI bracket that is vertically fixed to the first support member and the second support member, two ends of the PCI bracket away from the first support member and the second support member are provided with third fixing holes, the computer case body is provided with fourth fixing boles that match with the third fixing holes.

In some embodiments of the present disclosure, the PCI bracket is provided with at least two PCI card slots, the PCI bracket is further provided with at least two first connection holes, the PCI card slot is provided with second connection holes, and number and positions of first connection holes are correspond to those of the second connection holes.

In some embodiments of the present disclosure, the computer case body is provided with at least one transparent side panel.

The beneficial effects of the present disclosure are reflected in the stable fixation of the graphics card in the computer case through the setting of the graphics card holder and fixing member, and the vertical placement of the graphics card adapter cable, which enhances the heat dissipation performance of the graphics card and renders the colorful and cool fans visible intuitively, thereby solving problems of poor heat dissipation performance, unstable graphics card fixation, and low visual aesthetics in most computer case.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings are used to provide further understanding of the present disclosure and form a part of the specification. They are used together with the embodiments of the present disclosure to explain the present disclosure and do not constitute limitations on the present disclosure.

FIG. **1** is a perspective view of a graphics card holder provided by the present disclosure.

FIG. **2** is an exploded view of FIG. **1**.

FIG. **3** is a perspective view of a computer case body provided by the present disclosure.

FIG. **4** is a perspective view of the graphics card holder installed on the computer case body.

FIG. **5** is a perspective view of a graphics card vertically placed on the computer case with a graphics card holder.

FIG. **6** is a perspective view of a computer case with the graphics card holder.

Numeral reference: **1**—computer case body; **11**—second fixing hole; **12**—fourth fixing hole; **2**—graphics card holder; **21**—first support member; **211**—fixing member; **2111**—sliding hole; **2112**—graphics card fixing hole; **212**—threading port; **213**—sliding grooves; **22**—second support member; **221**—graphics card adapter cable; **222**—PCI bracket; **2221**—third fixing hole; **2222**—first connection hole; **223**—

PCI card slot; **2231**—second connection hole; **23**—third support member; **231**—first fixing hole.

DESCRIPTION OF EMBODIMENTS

The following will provide a clear and complete description of the technical solution in the embodiments of the present application, combined with the accompanying drawings. Obviously, the described embodiments are only a part of the embodiments of the present application, not all of them. Based on the embodiments in this application, all other embodiments those skilled in the art without creative work are within the protection scope of this application.

Referring to FIGS. **1** and **6**, a computer case with a graphics card holder includes a computer case body **1** and a graphics card holder **2**. The graphics card holder **2** includes a first support member **21**, a second support member **22**, and a third support member **23**. The first support member **21** is slidably connected to a fixing member **211**, the second support member **22** is vertically provided at one end of the first support member **21**, the second support member **22** is provided with a graphics card adapter cable **221**, and the third support member **23** is vertically provided on one side of the first support member **21**. One side of the third support member **23** away from the first support member **21** is fixedly connected to the computer case body **1**.

As shown in FIG. **2**, the graphics card holder **2** is integrally formed and bent to form the first support member **21**, the second support member **22**, and the third support member **23**. The graphics card adapter cable **221** is fixed to one end of the second support member **22** and connected to the graphics card bracket, and the other end thereof is connected to the graphics card slot on a computer case motherboard. In an implementation mode, the third support member **23** is vertically provided on one side of the first support member **21** away from the second support member **22**.

As shown in FIG. **2**, in some embodiments, the second support member **22** is a right-angled trapezoidal structure, and one right-angled side of the second support member **22** is connected to the first support member **21**.

As shown in FIGS. **1** and **2**, the right-angled trapezoidal structure of the second support member **22** ensures that the second support member **22** supports the graphics card through the graphics card adapter cable **221** while reducing the space occupied by the second support member **22** in the computer case body **1**.

In some embodiments, a threading port **212** with a size larger than the graphics card adapter cable **221** is provided on one side of the first support member **21** near the second support member **22**.

One end of the graphics card adapter cable **221** is fixed to the second support member **22**, and the other end thereof is connected to the graphics card slot on the computer case motherboard through the threading port **212**. The setting of the threading port **212** renders it more convenient for the graphics card adapter cable **221** to pass through the first support member **21** without extending a length of the graphics card adapter cable **221**. The setting of the threading port **212** and the graphics card adapter cable **221** enables the first support member **21** to separate the graphics card from the computer case motherboard without affecting the connection between the graphics card and the computer case motherboard, thereby achieving isolation of the main heating function modules such as the CPU on the graphics card and the computer case motherboard, and achieving better heat dissipation effect for each of the main heating function modules such as the graphics card and CPU.

In some embodiments, the fixing component **211** is integrally formed and bent to form a right angle. At least two sliding boles **2111** are provided on one side of the fixing member **211**, and a graphics card fixing hole **2112** is provided on the other side. In an implementation mode, there are three vertically spaced sliding holes **2111**, and the graphics card fixing bole **2112** is a vertically arranged elongated hole.

In some embodiments, at least two parallel sliding grooves **213** are provided on one side of the first support member **21** near the third support member **23**, and number and positions of the sliding grooves **213** are corresponding to those of the sliding holes **2111**. In an implementation mode, there are three vertically arranged sliding groove **213** with equal spacing. The fixing member **211** is fixed to the first support member **21** through a bolt that passes through the sliding holes **2111** and the sliding grooves **213**. The fixing member **211** can be flexibly on the first support member **21**, the fixing member **211** can be adjusted in position along the sliding grooves **213** on the first support member **21** to match graphics cards of different sizes. The setting of multiple sliding grooves **213** and the sliding holes **2111** renders the sliding of the fixing member **211** smoother, and a connection between the fixing member **211** and the first support member **21** more secure, thereby rendering the fixing member **211** more stably support the graphics card.

In some embodiments, a distance between the first support member **21** and the computer case motherboard is 4 cm to 8 cm. The computer case motherboard is mounted on the computer case body **1**, and the distance between the first support member **21** and the computer case motherboard ensures that the computer case motherboard has a larger heat dissipation space, achieving better heat dissipation effect and thus improving the performance of the computer case motherboard.

In some embodiments, one side of the third support member **23** away from the first support member **21** is provided with a first fixing hole **231**, and the computer case body **1** is provided with a second fixing hole **11** corresponding to the first fixing hole **231**. By the bolt passing through first fixing hole **231** and the second fixing bole **11**, the third support member **23** is fixedly connected to the computer case body **1**, thereby fixing the graphics card holder **2** to the computer case body **1**.

In some embodiments, one side of the second support member **22** is bent to form a PCI bracket **222** vertically fixed to the first support member **21** and the second support member **22**. Two ends of the PCI bracket **222** away from the first support member **21** and the second support member **22** are provided with third fixing holes **2221**, and the computer case body **1** is provided with fourth fixing holes **12** that match with the third fixing holes **2221**. By threading the third fixing holes **2221** and the fourth fixing boles **12** with bolts, the PCI bracket **222** is fixed to the computer case body **1**, thereby better fixing the graphics card holder **2** to the computer case body **1**.

In some embodiments, the PCI bracket **222** is provided with at least two PCI card slots **223**, the PCI bracket **222** is provided with at least two first connection holes **2222**, and the PCI card slot **223** is provided with second connection holes **2231**. The number and positions of the first connection holes **2222** are corresponded to those of the second connection holes **2231**. In an implementation mode, the PCI bracket **222** is provided with four first connection holes **2222**, and there are four PCI card slots **223**. Bach PCI card slot **223** is provided with one second connection hole **2231**. The PCI card slot **223** is fixed to the PCI bracket **222** by threading bolts through the first connection holes **2222** and the second connection holes **2231**.

In some implementations, the computer case body **1** is provided with at least one transparent side panel. After the graphics card is placed vertically, the colorful and cool fan on one side of the graphics card is visually visible through the transparent side panel, thereby rendering the product have a sense of technology. The colorful and cool fan is more likely to attract the attention and purchasing desire of the public, indirectly increasing product sales.

Working method, in an initial state, there is no graphics card holder **2** installed on the computer case. The graphics card holder **2** needs to be installed on the computer case body **1** first. First, the first fixing hole **231** and the second fixing hole **11** are threaded with bolts to fix the third support member **23** on the computer case body **1**. Then, the third fixing hole **2221** and the fourth fixing hole **12** are threaded with bolts to fix the PCI bracket **222** on the computer case body **1**. Finally, the PCI card slot **223** is fixedly connected to the PCI bracket **222** through the first connection holes **2222** and the second connection holes **2231**, thereby completing an installation of the graphics card holder **2**.

Finally, one side of the graphics card is placed with a fan facing away from the motherboard of the computer case vertically on the second support member **22**, so that one end of the graphics card adapter cable **221** fixed on the second support member **22** is connected to the graphics card bracket, and adjust the position of the fixing piece **211** on the sliding groove **213**, so that one side of the fixing member **211** provided with the graphics card fixing hole **2112** is tightly attached to the side of the graphics card, and then bolts are used to thread the graphics card fixing hole **2112** and the corresponding hole on the graphics card to fix and connect the graphics card to the fixing member **211**; the other end of the graphics card adapter cable **221** is connected through the threading port **212** to the graphics card slot on the computer case motherboard, thereby completing an installation of the graphics card.

In a description of the embodiments of the present disclosure, it should be understood that orientation or position relationship indicated with terms "up", "down", "front", "back", "left", "right", "straight", "horizontal", "center", "top", are based on the orientation or position relationship shown in the accompanying drawings, only for a convenience of describing the present disclosure and simplifying the description, and does not indicate or imply that the device or component referred to must have a specific orientation, be constructed and operated in a specific orientation, and therefore cannot be understood as a limitation of the present disclosure. Where "inner side" refers to the internal or enclosed area or space. "Peripheral" refers to the area around a specific component or region.

In the description of the embodiments of the present disclosure, terms "first", "second", "third", and "fourth" are only used for a descriptive purpose and cannot be understood as indicating or implying relative importance or implying the number of technical features indicated. Therefore, features that are limited to "first", "second", "third", and "fourth" may explicitly or implicitly include one or more of these features. In the description of the present disclosure, unless otherwise specified, the meaning of "multiple" refers to two or more.

In the description of the embodiments of the present disclosure, it should be noted that unless otherwise specified and limited, terms "installation", "connection to", "connection with", and "assembly" should be broadly understood, for example, it can be a fixed connection, a detachable connection, or an integrated connection; it can be directly connected, indirectly connected through an intermediate medium, or connected internally between two components. For those skilled in the art, specific meanings of the above terms in the present disclosure can be understood in specific situations.

Although the embodiments of the present disclosure have been shown and described, it will be understood by those skilled in the art that various changes, modifications, substitutions, and variations can be made to these embodiments without departing from the principles and spirit of the present disclosure. The scope of the present disclosure is limited by the appended claims and their equivalents.

What is claimed is:

1. A computer case with a graphics card holder, comprising a computer case body and a graphics card holder,

wherein the graphics card holder comprises a first support member, a second support member, and a third support member;

the first support member is slidably connected to a fixing member, the second support member is vertically provided at one end of the first support member, the second support member is provided with a graphics card adapter cable, the third support member is vertically provided on one side of the first support member, and one side of the third support member away from the first support member is fixedly connected to the computer case body;

wherein the fixing member is integrally formed and bent to form a right angle, at least two sliding holes are provided on one side of the fixing member, and a graphics card fixing hole is provided on the other side;

wherein at least two parallel sliding grooves are provided on one side of the first support member near the third support member, and number and positions of the sliding grooves are corresponding to those of the sliding holes.

2. The computer case with a graphics card holder according to claim **1**, wherein the second support member is a right-angled trapezoidal structure, and one right-angled side of the second support member is connected to the first support member.

3. The computer case with a graphics card holder according to claim **2**, wherein one side of the second support member is bent to form a PCI bracket that is vertically fixed to the first support member and the second support member,

two ends of the PCI bracket away from the first support member and the second support member are provided with third fixing holes,

the computer case body is provided with fourth fixing holes that match with the third fixing holes.

4. The computer case with a graphics card holder according to claim **3**, wherein the PCI bracket is provided with at least two PCI card slots, the PCI bracket is further provided with at least two first connection holes,

the PCI card slot is provided with second connection holes, and number and positions of first connection holes are correspond to those of the second connection holes.

5. The computer case with a graphics card holder according to claim **1**, wherein a threading port with a size larger than the graphics card adapter cable is provided on one side of the first support member near the second support member.

6. The computer case with a graphics card holder according to claim **1**, wherein a distance between the first support member and a computer case motherboard is 4 cm to 8 cm.

7. The computer case with a graphics card holder according to claim **1**, wherein one side of the third support member away from the first support member is provided with a first fixing hole, and the computer case body is provided with a second fixing hole corresponding to the first fixing hole.

8. The computer case with a graphics card holder according to claim **1**, wherein the computer case body is provided with at least one transparent side panel.

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